

Brewery Plat-Eau: Beer Production Optimisation

Optimisation Techniques E004120A

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1 Introduction

Brewery Plat-Eau is an optimisation project about choosing how to produce beer at minimum cost. The model decides how much barley, malt, malt extract, hops, and yeast to use for a requested beer volume and colour. Later tasks also require minimum ingredient quality.

The continuous decision variables are ingredient quantities. Two binary variables represent whether the malting and mashing processes are activated, because these processes have fixed costs. The report is organised as a direct answer to the assignment tasks: first the common brewing assumptions and optimisation model are defined, and then Tasks 1–7 are answered in order.

2 Problem Background and Brewing Assumptions

Only the brewing assumptions needed for the optimisation model are used. The fermentable material can be supplied through three routes: raw barley, purchased malt, and malt extract. Raw barley must first be converted to malt and then mashed. Purchased malt skips malting but still requires mashing. Malt extract is already in extract-equivalent form and can be used directly.

The final beer volume is related to wort volume by

$$V_{\text{beer}} = 0.95V_{\text{wort}}, \quad (2.1)$$

so that $V_{\text{wort}} = V_{\text{beer}}/0.95$. The required malt-extract-equivalent mass is

$$\frac{M_{\text{tot}}}{V_{\text{wort}}} = 2.9(SG - 1), \quad SG = 1.050. \quad (2.2)$$

The conversion rules for fermentable sources are

$$1 \text{ kg barley} = 0.75 \text{ kg malt}, \quad (2.3)$$

$$1 \text{ kg malt} = 0.8 \text{ kg malt extract equivalent}. \quad (2.4)$$

Therefore,

$$1 \text{ kg barley} = 0.6 \text{ kg malt extract equivalent}. \quad (2.5)$$

Hops are required in proportion to wort volume, and yeast is required in proportion to final beer volume:

$$M_{\text{hops}} = 0.0013V_{\text{wort}}, \quad (2.6)$$

$$M_{\text{yeast}} = 0.00075V_{\text{beer}}. \quad (2.7)$$

Beer colour is measured by EBC. Each colour class is represented by lower and upper EBC bounds, except black beer, which has only a lower bound. Quality is modelled as a weighted average within each ingredient group. All ingredient costs, EBC values, quality values, and process costs are read from `dataset_EN.txt`.

3 Model Setup

3.1 Decision Variables

Let B , M , E , H , and Y denote the sets of barley, malt, malt-extract, hop, and yeast alternatives. The continuous decision variables are

$$\begin{array}{lll}
 b_i \geq 0 & i \in B, & \text{kg of barley } i, \\
 m_i \geq 0 & i \in M, & \text{kg of malt } i, \\
 e_i \geq 0 & i \in E, & \text{kg extract-equivalent of extract } i, \\
 h_i \geq 0 & i \in H, & \text{kg of hops } i, \\
 y_i \geq 0 & i \in Y, & \text{kg of yeast } i.
 \end{array}$$

The binary process variables are

$$\begin{array}{ll}
 z_{\text{malt}} \in \{0, 1\}, & 1 \text{ if malting is used,} \\
 z_{\text{mash}} \in \{0, 1\}, & 1 \text{ if mashing is used.}
 \end{array}$$

For compact notation, the extract-equivalent coefficients are

$$a^B = 0.6, \quad a^M = 0.8, \quad a^E = 1. \quad (3.1)$$

3.2 Objective Function

The cost-minimisation objective includes ingredient costs, variable process costs, and fixed process costs:

$$\begin{aligned}
 \min C = & \sum_{i \in B} c_i^B b_i + \sum_{i \in M} c_i^M m_i + \sum_{i \in E} c_i^E e_i \\
 & + \sum_{i \in H} c_i^H h_i + \sum_{i \in Y} c_i^Y y_i \\
 & + v_{\text{malt}} \sum_{i \in B} b_i + v_{\text{mash}} \left(0.75 \sum_{i \in B} b_i + \sum_{i \in M} m_i \right) \\
 & + F_{\text{malt}} z_{\text{malt}} + F_{\text{mash}} z_{\text{mash}}.
 \end{aligned} \quad (3.2)$$

Here F denotes fixed process costs and v denotes variable process costs from the dataset.

3.3 Volume Constraints

For a requested beer volume, the wort volume and required extract-equivalent mass are computed using the conversion formulas in Section 2. The fermentable-source balance is

$$0.6 \sum_{i \in B} b_i + 0.8 \sum_{i \in M} m_i + \sum_{i \in E} e_i = M_{\text{tot}}. \quad (3.3)$$

The hop and yeast balances are

$$\sum_{i \in H} h_i = 0.0013 V_{\text{wort}}, \quad (3.4)$$

$$\sum_{i \in Y} y_i = 0.00075 V_{\text{beer}}. \quad (3.5)$$

3.4 Colour Constraints

The EBC formula is a weighted average over fermentable sources. Barley and malt are first converted to malt-extract-equivalent mass before the EBC average is formed. With $\gamma = (SG - 1)/0.0344$, define

$$\begin{aligned}
 \Phi_E = & \sum_{i \in B} E_i^B (0.6 b_i) + \sum_{i \in M} E_i^M (0.8 m_i) \\
 & + \sum_{i \in E} E_i^E e_i.
 \end{aligned} \quad (3.6)$$

The beer colour is

$$\text{EBC} = \gamma \frac{\Phi_E}{M_{\text{tot}}}. \quad (3.7)$$

For a target interval $[\text{EBC}_{\min}, \text{EBC}_{\max}]$, the linear constraints are

$$\gamma \Phi_E \geq \text{EBC}_{\min} M_{\text{tot}}, \quad (3.8)$$

$$\gamma \Phi_E \leq \text{EBC}_{\max} M_{\text{tot}}. \quad (3.9)$$

The assignment defines black beer as $\text{EBC} > 120$. Since LP solvers cannot model strict inequalities directly, black beer is modelled as

$$\text{EBC} \geq 120 + \epsilon_{\text{EBC}}, \quad \epsilon_{\text{EBC}} = 10^{-6}.$$

When $\text{EBC}_{\max} = \infty$, the upper EBC constraint is omitted.

3.5 Quality Constraints

Quality requirements are weighted averages rewritten as linear inequalities. For fermentable sources,

$$\sum_{i \in B} Q_i^B (0.6b_i) + \sum_{i \in M} Q_i^M (0.8m_i) + \sum_{i \in E} Q_i^E e_i \geq Q_{\min}^S M_{\text{tot}}. \quad (3.10)$$

For hops and yeast,

$$\sum_{i \in H} Q_i^H h_i \geq Q_{\min}^H \sum_{i \in H} h_i, \quad (3.11)$$

$$\sum_{i \in Y} Q_i^Y y_i \geq Q_{\min}^Y \sum_{i \in Y} y_i. \quad (3.12)$$

3.6 Process Constraints

Big-M constraints link ingredient use to the fixed-cost process variables. Barley requires malting, barley and purchased malt require mashing, and malting implies mashing:

$$\sum_{i \in B} b_i \leq M_{\text{malt}} z_{\text{malt}}, \quad (3.13)$$

$$0.75 \sum_{i \in B} b_i + \sum_{i \in M} m_i \leq M_{\text{mash}} z_{\text{mash}}, \quad (3.14)$$

$$z_{\text{malt}} \leq z_{\text{mash}}. \quad (3.15)$$

The implementation uses

$$M_{\text{malt}} = \frac{M_{\text{tot}}}{0.6}, \quad M_{\text{mash}} = \frac{M_{\text{tot}}}{0.8}, \quad (3.16)$$

which are large enough to allow all-barley or all-malt solutions.

3.7 LP Relaxation and Brute-Force Solution

The LP relaxation replaces the binary restrictions

$$z_{\text{malt}}, z_{\text{mash}} \in \{0, 1\}$$

with continuous bounds

$$0 \leq z_{\text{malt}}, z_{\text{mash}} \leq 1.$$

The relaxation gives a lower bound for cost minimisation, but fractional process variables are not valid production decisions.

The original MILP is solved by brute force over the valid binary process choices

$$(z_{\text{malt}}, z_{\text{mash}}) \in \{(0, 0), (0, 1), (1, 1)\}. \quad (3.17)$$

For each fixed process choice, one LP is solved using `scipy.optimize.linprog`. Since the model has only two binary variables, this enumeration is exact and does not require a dedicated MILP solver.

4 Task 1: First Analysis

This section answers assignment Task 3.1 by first converting a fixed barley amount to beer volume and then comparing cost as a function of beer volume for the three fermentable-source routes.

4.1 Beer Volume from 25.44 kg Barley

Using (2.5) and (2.2), the beer volume obtainable from a barley mass M_B is

$$V_{\text{beer}} = 0.95 \frac{0.6M_B}{2.9(SG - 1)}. \quad (4.1)$$

For $M_B = 25.44$ kg, the extract-equivalent mass is 15.264 kg, the wort volume is 105.269 L, and the final beer volume is 100.006 L.

Table 4.1: Task 1 Barley-to-Beer Calculation

Quantity	Expression	Value
Barley mass	M_B	25.44 kg
Extract equivalent	$0.6M_B$	15.264 kg
Wort volume	$0.6M_B/[2.9(SG - 1)]$	105.269 L
Beer volume	$0.95V_{\text{wort}}$	100.006 L

4.2 Cost Functions for Separate Malt-Source Routes

The first cost analysis uses the cheapest available barley, malt, malt extract, hops, and yeast from the dataset. With V denoting final beer volume in litres, the separate-route cost functions are

$$C_{\text{barley}}(V) = 3482.000 + 0.397529V, \quad (4.2)$$

$$C_{\text{malt}}(V) = 1072.000 + 0.661836V, \quad (4.3)$$

$$C_{\text{extract}}(V) = 1.338757V. \quad (4.4)$$

The barley route has the largest fixed cost because both malting and mashing are activated. The malt route has only the mashing fixed cost. The malt-extract route has no process fixed cost but has the largest variable cost per litre.

Table 4.2 shows how these costs are split between fermentable sources, hops, yeast, and process costs for three representative volumes. Hops and yeast increase almost linearly with volume in all routes; the important route changes come from the trade-off between fixed process costs and fermentable-source variable costs.

Table 4.2: Task 1 Cost Breakdown for Separate Source Routes

Volume	Route	Source	Hops	Yeast	Var. malt	Var. mash	Fixed malt	Fixed mash	Total
50	Barley	0.903	2.319	2.726	4.961	8.967	2410.000	1072.000	3501.876
50	Malt	19.079	2.319	2.726	0.000	8.967	0.000	1072.000	1105.092
50	Extract	61.892	2.319	2.726	0.000	0.000	0.000	0.000	66.938
5000	Barley	90.307	231.947	272.625	496.053	896.711	2410.000	1072.000	5469.643
5000	Malt	1907.895	231.947	272.625	0.000	896.711	0.000	1072.000	4381.178
5000	Extract	6189.211	231.947	272.625	0.000	0.000	0.000	0.000	6693.783
25000	Barley	451.535	1159.737	1363.125	2480.263	4483.553	2410.000	1072.000	13420.213
25000	Malt	9539.474	1159.737	1363.125	0.000	4483.553	0.000	1072.000	17617.888
25000	Extract	30946.053	1159.737	1363.125	0.000	0.000	0.000	0.000	33468.914

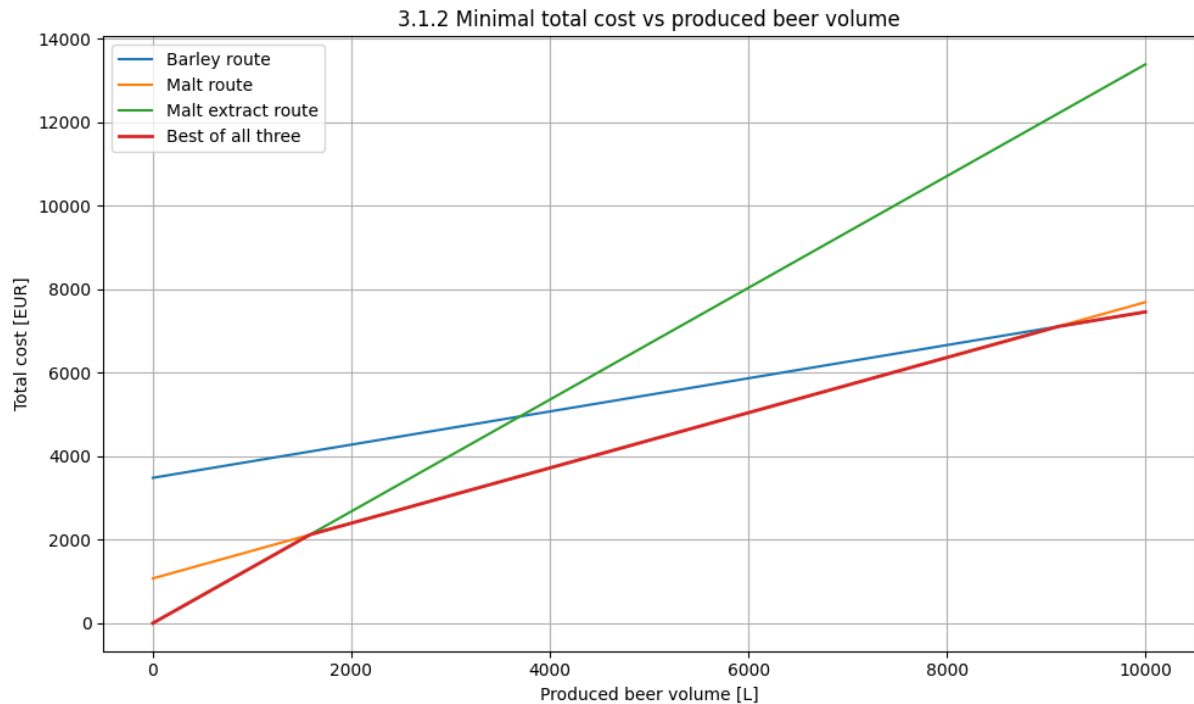


Figure 4.1: Task 1 cost as a function of beer volume for the barley, malt, malt extract, and combined minimum-cost routes.

4.3 Combined Malt-Source Solution

When all three source routes can be chosen, the combined minimum-cost curve is the lower envelope of the three separate cost curves. For small volumes, malt extract is cheapest because it avoids fixed process costs. For medium volumes, purchased malt becomes cheaper. For large volumes, raw barley becomes attractive because its variable cost per litre is lowest, even though it has the highest fixed process cost.

5 Task 2: LP and MILP Formulation

This section answers assignment Task 3.2. The full mathematical model is given once in Section 3; this task explains how it is used computationally.

5.1 MILP Formulation

The model is a mixed-integer linear program because all balances, colour constraints, quality constraints, and process-linking constraints are linear, while z_{malt} and z_{mash} are binary. The two binary variables determine whether fixed malting and mashing costs are charged.

5.2 LP Relaxation

The relaxed model uses the same equations as the MILP but replaces the binary restrictions by

$$0 \leq z_{\text{malt}}, z_{\text{mash}} \leq 1. \quad (5.1)$$

For cost minimisation, this gives a lower bound on the MILP optimum. A relaxed solution may contain fractional process variables, so it is not necessarily a valid production plan.

5.3 Brute-Force Method

Enumerating the binary variables is exact in this project because there are only two process choices and only three valid combinations:

$$(z_{\text{malt}}, z_{\text{mash}}) \in \{(0, 0), (0, 1), (1, 1)\}.$$

For each combination, the binary variables are fixed and the remaining continuous LP is solved. The best feasible solution among these LPs is the original MILP solution.

5.4 Programmatic Implementation

The model was implemented in Python using the assignment input data. For each problem instance, the program constructs the objective vector, equality constraints, inequality constraints, right-hand sides, and variable bounds for the LP. The LP subproblems are solved with `scipy.optimize.linprog`. No advanced MILP solver is used; the binary process choices are handled by brute-force enumeration. Result tables report only non-zero decision variables rather than the full optimisation vector.

6 Task 3: 5000 Litre Amber Beer

6.1 Problem Setup

The requested production volume is 5000 L. The beer colour is amber, so the EBC bounds are

$$20 \leq \text{EBC} \leq 29.$$

No quality constraints are imposed in this task.

Table 6.1: Task 3 Problem Setup

Parameter	Value
Beer volume	5000 L
Colour	amber
EBC bounds	20 to 29
Quality constraints	not active

6.2 LP Relaxation Result

The LP relaxation gives a cost of 4497.687. In this case the relaxed process variables are already integral: $z_{\text{malt}} = 0$ and $z_{\text{mash}} = 1$. The achieved colour is $\text{EBC} = 29.000$.

Table 6.2: Task 3 LP Relaxation Result

Item	Result
Cost	4497.687
z_{malt}	0
z_{mash}	1
Achieved EBC	29.000

Table 6.3: Task 3 Non-Zero Ingredients in the Relaxed Solution

Group	Variable	Amount (kg)
Malt	<code>malt04</code>	685.349
Malt	<code>malt06</code>	268.598
Hops	<code>hop07</code>	6.842
Yeast	<code>yeast05</code>	3.750

6.3 Why Most Ingredients Are Unused

Most ingredients are unused because an LP optimum usually occurs at a corner point of the feasible region. The solver selects the cheapest combination that satisfies the active volume and colour constraints. Ingredients that are more expensive or have less useful EBC values receive zero mass.

6.4 Rounded Relaxation

The rounding step is applied only to the binary process variables. The relaxed solution gives $(z_{\text{malt}}, z_{\text{mash}}) = (0, 1)$, so the rounded process choice is also $(0, 1)$. The continuous LP is then re-solved with that process choice fixed. Directly rounding all ingredient quantities would be incorrect because ingredient amounts are continuous production decisions.

Table 6.4: Task 3 Rounded-Process Re-Solve

Item	Result
Rounded process choice	(0, 1)
Re-solved cost	4497.687
Achieved EBC	29.000
Ingredient mix	same as Table 6.3

6.5 Original MILP Result

The brute-force MILP solve checks the valid process choices (0, 0), (0, 1), and (1, 1). The optimal process choice is (0, 1), with total cost 4497.687 and achieved EBC = 29.000.

Table 6.5: Task 3 Original MILP Result

Item	Result
Selected process choice	(0, 1)
Total cost	4497.687
Achieved EBC	29.000
Ingredient mix	same as Table 6.3

7 Task 4: Less or More?

7.1 Results for 50 L and 25000 L

The amber-beer optimisation is repeated for 50 L, 5000 L, and 25000 L. The relaxation, rounded-process solution, and MILP solution have the same costs for these three cases because the relaxed process variables are already integral.

Table 7.1: Task 4 Volume Comparison

Beer volume (L)	Relaxation cost	Rounded cost	MILP cost	MILP cost per litre
50	67.767	67.767	67.767	1.355
5000	4497.687	4497.687	4497.687	0.900
25000	13420.213	13420.213	13420.213	0.537

The corresponding MILP process choices are (0, 0) for 50 L, (0, 1) for 5000 L, and (1, 1) for 25000 L. Thus, the smallest batch can be produced without activating malting or mashing, the 5000 L case uses mashing only, and the largest batch uses both malting and mashing.

The 5000 L ingredient solution is already reported in Task 3. Tables 7.2 and 7.3 therefore give the additional MILP solution details for 50 L and 25000 L. The quality values are reported for completeness, but they are not constrained in this task.

Table 7.2: Task 4 MILP Solution Summary for Additional Volumes

Volume (L)	Process	Cost	EBC	Q_S	Q_H	Q_Y
50	(0, 0)	67.767	29.000	1.000	1.000	1.000
25000	(1, 1)	13420.213	20.000	1.000	1.000	1.000

Table 7.3: Task 4 Non-Zero Ingredients for 50 L and 25000 L

Volume (L)	Group	Variable	Amount (kg)
50	Malt extract	<code>maltextract04</code>	0.721
50	Malt extract	<code>maltextract10</code>	6.911
50	Hops	<code>hop07</code>	0.068
50	Yeast	<code>yeast05</code>	0.037
25000	Barley	<code>barley04</code>	799.499
25000	Barley	<code>barley09</code>	5560.150
25000	Hops	<code>hop07</code>	34.211
25000	Yeast	<code>yeast05</code>	18.750

7.2 Marked Costs on Task 1 Plot

The three production volumes from Table 7.1 are marked on the cost-volume plot from Task 1.

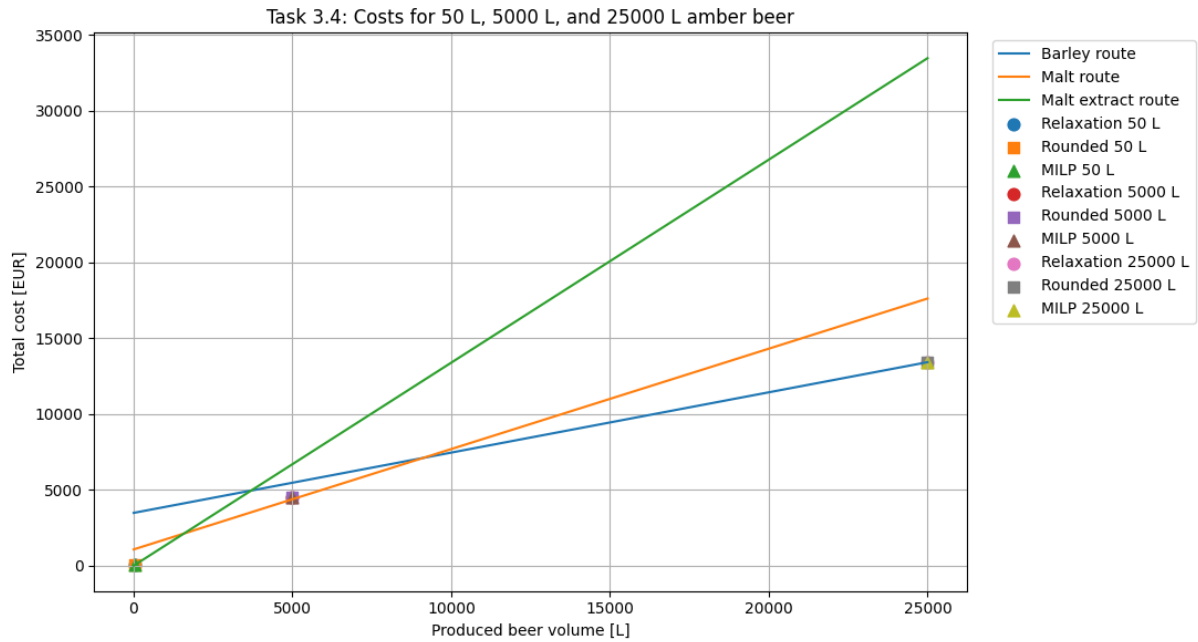


Figure 7.1: Task 4 costs for 50 L, 5000 L, and 25000 L marked on the Task 1 cost-volume plot.

The marked points follow the combined minimum-cost curve from Task 1. This is expected because the optimal source route changes with volume as fixed process costs become less important relative to variable ingredient costs.

7.3 Cost per Litre Discussion

The cost per litre decreases as the production volume increases. The 50 L batch is expensive per litre because any fixed process cost is spread over very few litres. Larger batches spread fixed costs over more output, so the cost per litre is lower. The 25000 L case has the lowest cost per litre in Table 7.1.

8 Task 5: Full Problem with Quality Constraints

8.1 Problem Setup

This task uses the same production volume and colour as Task 3, but adds minimum average quality constraints. The setup is 5000 L amber beer with

$$Q_{\min}^S = 3, \quad Q_{\min}^H = 3, \quad Q_{\min}^Y = 3.$$

Table 8.1: Task 5 Problem Setup

Parameter	Value
Beer volume	5000 L
Colour	amber
EBC bounds	20 to 29
Malt-source quality	at least 3
Hop quality	at least 3
Yeast quality	at least 3

8.2 Optimal Solution

The optimal quality-constrained solution uses process choice $(z_{\text{malt}}, z_{\text{mash}}) = (0, 1)$. The total cost is 4891.923 and the achieved EBC is 20.000. All three achieved group qualities are 3.000.

Table 8.2: Task 5 Optimal Solution Summary

Item	Result
Total cost	4891.923
Process choice	(0, 1)
Achieved EBC	20.000
Malt-source quality	3.000
Hop quality	3.000
Yeast quality	3.000

Table 8.3: Task 5 Non-Zero Ingredients

Group	Variable	Amount (kg)
Malt	malt03	901.457
Malt	malt06	17.497
Malt	malt12	34.994
Hops	hop07	3.421
Hops	hop10	3.421
Yeast	yeast03	3.750

8.3 Comparison with Task 3

The quality constraints increase the cost from 4497.687 in Task 3 to 4891.923 in Task 5. This is expected because the feasible region becomes smaller. The model can no longer use only the cheapest low-quality combination from Task 3; it must choose ingredients whose weighted averages satisfy the three quality lower bounds. The optimal EBC also moves from the upper amber boundary in Task 3 to the lower amber boundary in Task 5.

9 Task 6: From Light Blond to Black

9.1 Results for All Beer Colours

The full quality-constrained model from Task 5 is solved for each colour class. Black beer is modelled as $EBC \geq 120 + \epsilon_{EBC}$, with $\epsilon_{EBC} = 10^{-6}$, because the assignment specifies $EBC > 120$.

Table 9.1: Task 6 Results for All Beer Colours

Colour	EBC range	Feasible	Cost	Achieved EBC	Notes
light blond	6–8	no	–	–	No feasible quality-constrained mix found.
blond	9–12	yes	4919.601	12.000	Uses the upper blond boundary.
gold	13–19	yes	4883.000	14.535	Cheapest feasible colour in this run.
amber	20–29	yes	4891.923	20.000	Same setup as Task 5.
copper	30–45	yes	4908.251	30.000	Uses the lower colour boundary.
brown	46–75	yes	4934.376	46.000	Darker colour requires a different source mix.
dark brown	76–120	yes	4983.359	76.000	Upper bound remains 120.
black	> 120	yes	5061.830	120.000001	Modelled as $120 + \epsilon_{EBC}$.

Table 9.2 gives the selected process choice and achieved qualities for each feasible colour. Table 9.3 gives the corresponding non-zero ingredient mixes.

Table 9.2: Task 6 Solution Summary for Feasible Beer Colours

Colour	Process	Cost	EBC	Q_S	Q_H	Q_Y
blond	(0, 1)	4919.601	12.000	3.000	3.000	3.000
gold	(0, 1)	4883.000	14.535	3.000	3.000	3.000
amber	(0, 1)	4891.923	20.000	3.000	3.000	3.000
copper	(0, 1)	4908.251	30.000	3.000	3.000	3.000
brown	(0, 1)	4934.376	46.000	3.000	3.000	3.000
dark brown	(0, 1)	4983.359	76.000	3.000	3.000	3.000
black	(0, 1)	5061.830	120.000001	3.000	3.000	3.000

Table 9.3: Task 6 Non-Zero Ingredient Mixes for Feasible Colours

Colour	Non-zero ingredients
blond	malt03 454.842 kg; malt04 332.737 kg; malt11 166.368 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg
gold	malt03 953.947 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg
amber	malt03 901.457 kg; malt06 17.497 kg; malt12 34.994 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg
copper	malt03 805.411 kg; malt06 49.512 kg; malt12 99.024 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg
brown	malt03 651.737 kg; malt06 100.737 kg; malt12 201.474 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg
dark brown	malt03 363.598 kg; malt06 196.783 kg; malt12 393.566 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg
black	malt01 72.431 kg; malt06 293.839 kg; malt12 587.677 kg; hop07 3.421 kg; hop10 3.421 kg; yeast03 3.750 kg

9.2 Cost as a Function of Beer Colour

The minimum cost for each feasible colour is plotted in Figure 9.1.

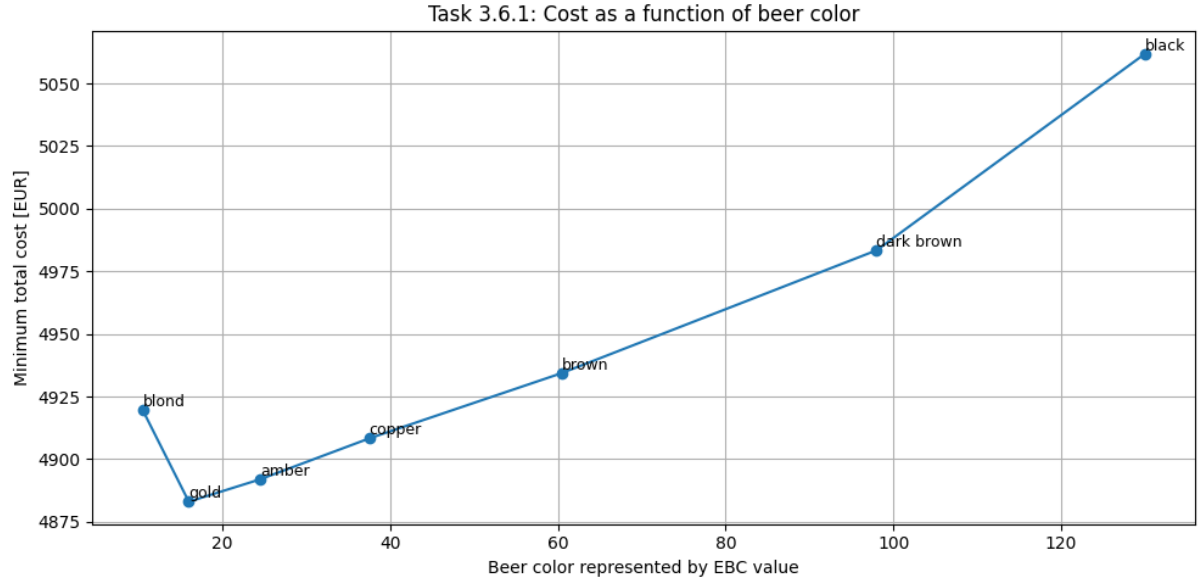


Figure 9.1: Task 6 minimum production cost for each feasible beer colour.

9.3 Explanation

The cost differences are caused by the available EBC values and quality scores in the dataset. Gold is the cheapest feasible colour in Table 9.1, while black is the most expensive feasible colour. All feasible colour classes use the same process choice, $(0, 1)$, so the main cost change comes from the fermentable-source mix rather than from different fixed process costs.

Light blond is infeasible because no available quality-constrained weighted EBC mix satisfies the low EBC interval 6–8. As the required colour becomes darker, the model introduces darker and more expensive malt sources to move the weighted EBC average upward. Black is feasible only by reaching just above the strict 120 EBC boundary, represented in the LP as $120 + \epsilon_{\text{EBC}}$.

10 Task 7: Maximising the Quality

10.1 New Objective

To maximise quality, introduce a scalar variable q representing the minimum achieved quality over the malt-source, hop, and yeast groups. The objective is

$$\max q, \quad (10.1)$$

or equivalently $\min -q$ in minimisation form.

10.2 Quality Constraints

The volume, colour, and process constraints remain active. For the 5000 L amber problem, the required totals are fixed constants:

$$M_{\text{tot}} = 763.158 \text{ kg}, \quad H_{\text{req}} = 6.842 \text{ kg}, \quad Y_{\text{req}} = 3.750 \text{ kg}. \quad (10.2)$$

The three group-quality constraints are therefore linear when rewritten with q as the lower bound:

$$\sum_{i \in B} Q_i^B (0.6b_i) + \sum_{i \in M} Q_i^M (0.8m_i) + \sum_{i \in E} Q_i^E e_i \geq qM_{\text{tot}}, \quad (10.3)$$

$$\sum_{i \in H} Q_i^H h_i \geq qH_{\text{req}}, \quad (10.4)$$

$$\sum_{i \in Y} Q_i^Y y_i \geq qY_{\text{req}}. \quad (10.5)$$

The terms qM_{tot} , qH_{req} , and qY_{req} are products of q with constants, not products of decision variables. The bound $0 \leq q \leq 5$ is added. The model remains a MILP because q is continuous and the only integer variables are still the two process variables handled by the same brute-force enumeration as in Section 3.7.

10.3 Can the Minimum Quality Be 5?

Yes. The optimisation reaches $q^* = 5.000$, so minimum quality 5 is feasible. Since 5 is also the upper bound on q , this is the best possible value. No ingredient group prevents quality 5 in the computed solution: the malt-source, hop, and yeast qualities all reach 5 while satisfying the amber colour constraint.

Table 10.1: Task 7 Quality Maximisation Result

Item	Result
Maximum minimum quality	5.000
Quality 5 feasible?	yes
Process choice	(0, 1)
Achieved EBC	20.000
Malt-source quality	5.000
Hop quality	5.000
Yeast quality	5.000

Table 10.2: Task 7 Non-Zero Ingredients

Group	Variable	Amount (kg)
Malt	<code>malt11</code>	870.772
Malt extract	<code>maltextract03</code>	66.541
Hops	<code>hop10</code>	6.842
Yeast	<code>yeast01</code>	3.750

11 Conclusion

The Brewery Plat-Eau tasks are solved with one common linear optimisation model and task-specific parameter choices. Fixed malting and mashing costs are represented with two binary variables. The LP relaxation gives a lower bound, while brute-force enumeration over the valid binary process choices gives the original MILP solution using only `scipy.optimize.linprog`.

The results show how volume, colour, and quality constraints change the cheapest production plan. Fixed process costs explain why small batches are more expensive per litre, and the EBC and quality constraints explain why some colours or ingredient mixes cost more than others.

References

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